Classical and Self-Supervised Hankel Filtering for Magnitude-Only Rydberg Channel Estimation

Abdullah Haatim

***Abstract*—Rydberg atomic receivers can estimate wireless channels from signal magnitudes measured with a known reference. Spatial channel structure can remove noise, but excessive filtering can also remove useful paths. We study this trade-off through classical estimation and a compact self-supervised estimator. The learned method keeps the exact Rician measurement update and uses a small controller to adjust smooth Hankel filtering. It learns from withheld pilot measurements rather than true channels. Paired simulations show when classical filtering helps and when it introduces bias. The compact estimator improves average accuracy over the tested smaller learned controls and offline-tuned OptShrink. Per-block rank-selected OptShrink is more accurate on the main average, but takes substantially longer in the tested implementation. Training with all pilots improves average accuracy at the same inference cost, yet increases severe errors. Mixed training and a simple selector do not resolve this trade-off. Weak references, nearly noiseless observations and wrong noise inputs remain limitations. The result is a measured accuracy–cost trade-off for a calibrated magnitude model, not universal superiority or a hardware demonstration.**

***Index Terms*—Channel estimation, Hankel matrices, magnitude measurements, Rydberg receivers, self-supervised learning.**

I. Introduction

RYDBERG atomic receivers sense radio-frequency electric fields through highly excited atoms. Their sensitivity and broad operating range motivate research on wireless communication and sensing, including future networks [1]. Array models extend this setting to multiple users [2]. In the magnitude-only receiver considered here, a known reference combines with the incoming field before amplitude measurement. The receiver does not directly observe complex baseband samples. Recovering the complex channel is therefore a nonlinear inverse problem, even when the pilots and reference are known.

A small number of propagation paths also gives the channel structure across the array. A Hankel matrix built from a uniform linear array channel has low rank. Filtering this matrix can suppress noise, but hard truncation can erase weak paths. The appropriate amount of filtering depends on the measurements and on channel complexity. Searching for a rank separately for every received block can be expensive. This motivates a practical question: can a small learned rule reuse offline training to provide useful structural correction at lower processing cost, and when does a well-tuned classical method remain better?

This paper combines three contributions within one observation model. First, paired classical experiments relate filtering gains to array size, effective rank and repeated projection. Second, a 188-parameter estimator learns smooth spatial filtering from withheld measurements while retaining exact Rician expectation–maximization (EM) updates. Third, matched comparisons measure accuracy, severe errors and processing time against compact learned controls and strong OptShrink adaptations. Separate training and noise-input checks test the limits of this trade-off. The contribution is a specific compact design and evidence about when it is useful, not a new low-rank principle or a universally better shrinker.

A. Relation to Existing Work

Cui et al. developed EM-based Gerchberg–Saxton (EM-GS) for atomic-MIMO detection [2]; we apply it to known-pilot channel estimation. Unlike strong-reference linearization [3], we retain exact magnitudes. URformer already unfolds Rydberg phase retrieval [4]; our results do not reproduce its published performance. Quantum-MUSIC and its robust variant address direction finding and outliers [5], [6], not the estimation trade-off studied here.

Classical structure processing includes Cadzow's alternating projections [7], Hankel recovery [8], low-rank Hankel denoising (LRHD) [9], and structured alternating least squares [10]. OptShrink chooses data-driven singular-value weights [11]. Variational Bayesian low-rank phase retrieval [12] and Vandermonde factorization [13] offer other priors. Their original observations and guarantees differ from referenced Rician recovery.

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Our in-loop OptShrink adaptation is a strong control, not a reproduction of full LRHD or these other methods. Their guarantees are not transferred to the dependent errors produced by nonlinear reconstruction and Hankel lifting.

Compact learned estimators also have clear precedents. C-TISTA uses four scalars per stage for complex inverse problems [14]. ALISTA retains analytic weights and learns steps and thresholds [15]; OAMP-Net2 uses four scalars per stage for MIMO detection [16]. NA-ALISTA uses recurrence and observed residual summaries to control sparse recovery [17]. Compactness and input-dependent control are not new by themselves. Learned Hankel reconstruction [18], withheld-measurement self-supervision [19], [20], Rician-likelihood learning [21], and self-supervised Rydberg-signal denoising [22] also precede this work. Our specific design adjusts smooth spatial Hankel filtering inside exact referenced-magnitude EM. The matched recurrent control tests one adaptation of that broader idea, not superiority over published NA-ALISTA.

II. System Model and Available Information

Figure 1 shows the receiver. An N-element array receives P pilots from K users:

$$Z = |GS + B + W|, \qquad W_{np} \sim \mathcal{CN}(0, v). \quad (1)$$

Here $G \in \mathbb{C}^{N\times K}$ is the unknown channel, $S \in \mathbb{C}^{K\times P}$ contains known pilots, and B is the known reference field. Magnitude acts entrywise. Noise samples are independent, with variance $v/2$ in each real component. Receiver conversion gain is absorbed into the scaling. This effective model does not simulate atomic dynamics, optical readout or hardware drift.

For a half-wavelength uniform linear array, user k has channel

$$g_{nk} = \sum_{\ell=1}^{L_k} \alpha_{\ell k} e^{-j\pi n \sin\theta_{\ell k}}, \quad n = 0, \dots, N-1. \quad (2)$$

We draw $\alpha_{\ell k} \sim \mathcal{CN}(0, 1/L_k)$ and independent angles uniformly on $(-\pi/2, \pi/2)$. Pilots follow $\mathcal{CN}(0,1)$. The reference is broadside with constant waveform and zero phase:

$$B_{np} = 10^{\mathrm{RSR}/20}, \qquad v = K10^{-\mathrm{SNR}/10}. \quad (3)$$

SNR is relative to aggregate user power; reference-to-signal ratio (RSR) is relative to one user's power. At $K = 3$, RSR 12 dB is 7.23 dB relative to total signal power. Each user's expected channel power is one; individual realizations are not normalized to identical energy. Main comparisons supply the correct reference and noise variance to every estimator.

A. Likelihood, Ambiguity and Local Recovery

For $a = |\mu|$, a magnitude observation has Rician density

$$p(z \mid a, v) = \frac{2z}{v} e^{-(z^2+a^2)/v} I_0(2za/v), \quad z \ge 0. \quad (4)$$

For fixed v, its negative log-likelihood, up to observation-only terms, is

$$\ell(z, \mu, v) = \frac{(z - |\mu|)^2}{v} - \log I_{0e}(2z|\mu|/v), \quad (5)$$

where $I_{0e}(x) = e^{-x} I_0(x)$ for $x \ge 0$. Scaled Bessel functions avoid overflow. If variance is estimated, the omitted $\log v$ term must be restored.

For one array row, write $\mu_p = g^T s_p + b_p$ and use real coordinates $x = [\mathrm{Re}\, g^T, \mathrm{Im}\, g^T]^T$. At nonzero μ_p, the noiseless magnitude Jacobian has row

$$A_p = [\mathrm{Re}\, c_p^T, -\mathrm{Im}\, c_p^T], \qquad c_p = \mu_p^* s_p / |\mu_p|. \quad (6)$$

If A has rank $2K$, an invertible square sub-Jacobian gives local recovery by the inverse function theorem. Thus $P \ge 2K$ is necessary for this regular full-rank condition, but not sufficient. A small minimum singular value also makes recovery sensitive to noise. This is a local statement for an unrestricted row, not a global or sparse-channel recovery theorem [23].

With $B = 0$, multiplying g by a common phase leaves its magnitudes unchanged. A nonzero reference can break this ambiguity, but not for every design. If $b_p = c^T s_p$, rotating $g + c$ still preserves the observations. An unknown common reference phase also allows simultaneous rotation of channel and reference. Absolute phase cannot then be recovered without another phase anchor. Known reference calibration is a substantive assumption, not just a choice of units.

B. Information References

For independent measurements, the real Fisher information is

$$J = A^T \,\mathrm{diag}(w_p) A,$$
$$w_p = \mathbb{E}\left[\left(-\frac{2a_p}{v} + \frac{2Z_p}{v}\frac{I_1(2Z_p a_p/v)}{I_0(2Z_p a_p/v)}\right)^2\right]. \quad (7)$$

One magnitude contributes at most one information direction. The full channel information combines array rows in the same real-coordinate ordering. Where the required matrices are nonsingular, the unrestricted reference is J^{-1}. The geometric constrained Cramér–Rao bound (CCRB) is [24]

$$C_{\mathrm{geom}} = U(U^T J U)^{-1} U^T, \quad (8)$$

where U spans the real tangent space at the true path configuration. With $\omega = \pi \sin\theta$, its complex derivative

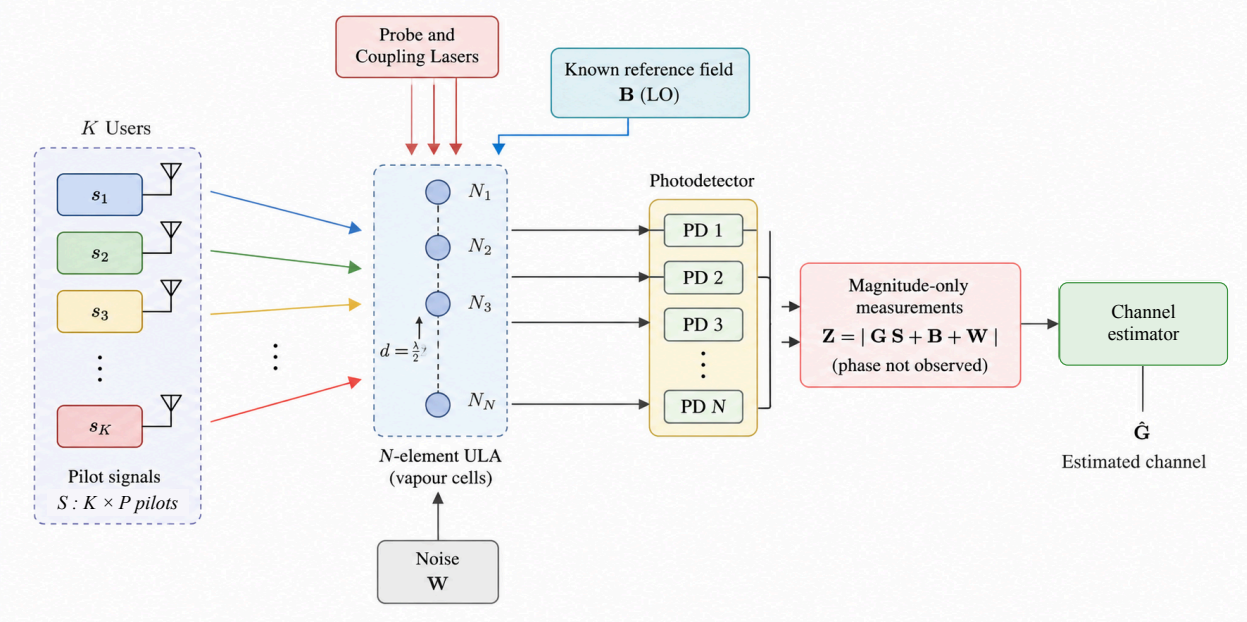


Fig. 1. Uplink receiver model. K users send pilots to N vapour-cell elements in a half-wavelength uniform linear array. A known RF reference combines with the received field. The optical readout supplies magnitude-only measurements to a channel estimator, covering both classical and learned methods. Lasers and photodetectors are schematic; atomic dynamics and optical readout are not simulated.

columns are $e^{-jn\omega}$, $je^{-jn\omega}$ and $-jn\alpha e^{-jn\omega}$. Real and imaginary stacking gives the tangent basis. True path count is used only for this ideal diagnostic.

We normalize summed traces by summed channel energy. The gap between these references describes the local value of geometric information under unbiasedness assumptions. It is not guaranteed attainable improvement. Our filtered estimators can be biased, so these curves are not absolute lower limits on their measured error.

III. Classical Structural Estimation

A. Exact Rician Measurement Update

Let $M = GS + B$ and $\kappa = 2Z|M|/v$. Conditional averaging of the unobserved complex field gives

$$Q = Z \odot \frac{I_1(\kappa)}{I_0(\kappa)} \odot \frac{M}{|M|},$$
$$G_{\mathrm{EM}} = (Q - B)S^H(SS^H)^{-1}. \tag{9}$$

Ratios act entrywise; the first line has continuous limit zero at $M = 0$. A linear solve evaluates the second line, requiring a full-row-rank fitting pilot matrix. For $\mathcal{L}(G) = \sum_{np} \ell(Z_{np}, M_{np}, v)$,

$$D = \frac{\partial \mathcal{L}}{\partial G^*} = \frac{(M - Q)S^H}{v},$$
$$G_{\mathrm{EM}} = G - vD(SS^H)^{-1}. \tag{10}$$

EM is therefore an exact preconditioned gradient update for this likelihood. The packed real gradient is $2D$; this factor matters when comparing gradient-based and EM-based stages.

All iterative comparisons use spectral initialization. For row n, form C_n with rows $[s_p^T, b_{np}]$ and take the leading eigenvector u_n of $C_n^H \operatorname{diag}(z_n) C_n$. Set

$$c_n = \frac{\sum_p z_{np} |(C_n u_n)_p|}{\sum_p |(C_n u_n)_p|^2}. \tag{11}$$

The first K entries of $c_n u_n e^{-j \arg(u_{n,K+1})}$ initialize the channel row. When observations are withheld, only the fitting subset enters initialization. Otherwise the estimator would contain information from its validation observations.

B. Hankel Lifting and Rank Selection

Let $q = \lceil N/2 \rceil$, $m = N - q + 1$, and

$$[\mathcal{H}(g)]_{ij} = g_{i+j}, \quad 0 \le i < m, \quad 0 \le j < q. \tag{12}$$

A channel with L distinct spatial frequencies has Hankel rank at most $\min(L, m, q)$. The inverse map $\mathcal{H}^\dagger$ averages antidiagonals and satisfies $\mathcal{H}^\dagger \mathcal{H}(g) = g$. A hard-rank sweep retains the largest r singular components and then averages antidiagonals. Repeating these operations is a Cadzow procedure; finitely many sweeps do not give an exact projection onto the intersection of Hankel and rank constraints.

Our Hankel-structured GS (HS-GS) baseline applies these sweeps after each EM update. It fits candidate

shared user ranks $r = 1, \dots, q$ on the first 70% of pilots and selects the smallest held-out squared amplitude residual on the remainder. It then restarts on all pilots. Unless specified otherwise, candidate/final budgets are 25/100 EM updates and each update uses four sweeps. Rank q explicitly bypasses filtering and recovers the paired EM-GS output. This option does not guarantee that validation chooses the bypass whenever filtering is harmful. We call amplitude-loss selection HS-amplitude and Rician-loss selection HS-Rician. Shared rank is a computational choice, not an assumption of equal user path counts.

C. Smooth Filtering and OptShrink Controls

The smooth operation replaces each singular value by

$$\widehat{\sigma}_i = \frac{\sigma_i^3}{\sigma_i^2 + \tau^2}. \tag{13}$$

A classical control selects τ as a multiple of the current RMS singular value, using held-out amplitude error. Multipliers are $\{0, 0.05, 0.1, 0.2, 0.4, 0.8, 1.6\}$; zero bypasses filtering. Update and selection budgets match HS-amplitude. This empirical rule is not OptShrink and has no claimed statistical optimality.

OptShrink instead estimates retained weights from the discarded spectrum [11]. For singular values $s_1 \geq \cdots \geq s_q$ and candidate rank $r < q$, define

$$\begin{aligned} \phi_d(z) &= \frac{1}{d-r}\left[\sum_{j=r+1}^{q} \frac{z}{z^2 - s_j^2} + \frac{d-q}{z}\right], \\ \mathcal{D}(z) &= \phi_m(z)\phi_q(z), \quad \widehat{s}_i = -2\mathcal{D}(s_i)/\mathcal{D}'(s_i),\ i \leq r. \end{aligned} \tag{14}$$

Other weights are zero. We rescale the spectrum, guard unresolved poles, average antidiagonals and mix the result with the measurement update. The adaptation is applied either after every update or once at the end.

The broad offline search uses the learned estimator's 2,048 validation observations and the same 21/9 masks. It covers both placements, ranks 1–15 plus no filtering, mixing factors $0.25, 0.5, 0.75, 1$, and update budgets $5, 10, 20, 50, 100$: 610 combinations. Rician validation loss is averaged equally over six path regimes. Settings are frozen for each of 21 noise/reference pairs before testing. Path labels balance offline selection; they are not inference inputs. Off-grid stress tests use the nearest lookup without retuning.

A stronger in-loop control selects rank for each received block. It keeps the lookup's mixing and iteration budget, fits ranks 1–15 plus bypass to 21 pilots, scores nine held-out pilots with Rician loss, and restarts on all 30. All search and refitting time counts. This is not an unrestricted search over user-specific ranks, mixing factors and stopping times. It uses no extra pilot observations, but pays computation to use the current block rather than only offline averages.

IV. Compact Self-Supervised Estimator

A. Measurement and Structural Stages

The primary estimator uses ten stages. At stage t, it relaxes the exact EM update:

$$G_{\mathrm{dc}} = G + \gamma_t(G_{\mathrm{EM}} - G), \qquad \gamma_t = 2\,\mathrm{sigmoid}(a_t). \tag{15}$$

The ten learned a_t start at $\gamma_t = 1$. For user k, write $H_k = \mathcal{H}(g_{\mathrm{dc},k})$, $s_H = \|H_k\|_F/\sqrt{q}$, and $p_i = \sigma_i/\sum_j \sigma_j$. A shared 8–16–2 network with SiLU activation receives

$$\begin{aligned} f_1 &= \exp(-\textstyle\sum_i p_i \log p_i)/q, \quad f_2 = p_1, \\ f_3 &= \textstyle\sum_{i=q-\lceil q/2\rceil+1}^{q} p_i, \\ f_4 &= \log\!\big(1 + \mathrm{mean}[(Z - |G_{\mathrm{dc}}S + B|)^2]/v\big)\,, \\ f_{5,k} &= \log\!\big(1 + v\,\mathrm{tr}[(SS^H)^{-1}]/(K\widehat{p}_k)\big)\,, \end{aligned} \tag{16}$$

and $P/(2K)$, $\log q$, $(t+1)/10$. Here $\widehat{p}_k = \|g_{\mathrm{dc},k}\|_2^2/N$, $t = 0, \dots, 9$, and the residual mean uses all fitting observations. These inputs describe the current spectrum, residual, noise, pilot support, size and stage. They contain no true channel, true rank or generator label.

Outputs (u, w) set $\tau = s_H[0.001 + \mathrm{softplus}(u)]$ and $\beta = \mathrm{sigmoid}(w)$. The smooth filter and channel update are

$$F_\tau(H) = H - \tau^2 H(H^H H + \tau^2 I)^{-1}, \tag{17}$$

$$g_k^{t+1} = g_{\mathrm{dc},k} + \beta_{t,k}[\mathcal{H}^\dagger F_{\tau_{t,k}}(H_k) - g_{\mathrm{dc},k}]. \tag{18}$$

The solve implements (13) after normalization by s_H. Gradients pass through it without an identity-backward approximation. Spectral features still require singular values. Threshold and mixing start near $0.1 s_H$ and 0.1. The network has 178 parameters; ten EM steps bring the total to 188.

For reproducibility, channel power and singular-value sum are floored at 10^{-12}, and squared Hankel RMS scale at 10^{-24}. Pilot-Gram systems use direct solves without an added ridge. PyTorch implementations use scaled Bessel functions and a series for $\kappa^2 < 10^{-4}$. OptShrink zeros weights at relative squared spectral gaps at most 10^{-13}. These are fixed numerical rules, not tuned extra physical assumptions. Positive threshold makes the reconstruction solve smooth, although ordered spectral features need not be differentiable at every tie.

Threshold and gate can compensate for each other. The gate is often near one in the fitted models; most adaptation changes filtering strength rather than providing an on/off decision. Neither output is an estimated physical rank. We measure actual correction by $\|g_k^{t+1} - g_{\mathrm{dc},k}\|_2/(\|g_{\mathrm{dc},k}\|_2 + \epsilon)$ rather than interpreting the gate alone as reliable abstention.

B. Learning Without True-Channel Targets

Each training block randomly partitions 30 pilots into 21 context columns A and nine withheld columns V. Only context data enter initialization, features and updates. The loss is

$$\mathcal{L}_{\text{hold}} = \frac{1}{N|V|} \sum_{n,p\in V} \ell\big(Z_{np}, [\widehat{G}_A S_V + B_V]_{np}, v\big). \quad (19)$$

Masks change every epoch. Independent validation blocks use fixed masks. Checkpoint selection minimizes the equally weighted mean held-out likelihood across validation conditions, including the untrained checkpoint. No channel targets choose weights or checkpoints. At inference, weights remain fixed and all pilots enter the estimator. There is no per-block training or rank search.

Training on 21 input pilots and testing on 30 changes the input distribution. Held-out likelihood measures prediction quality, not channel NMSE. Our original withheld-pilot estimator remains the main method.

As a separate training check, we use all 30 pilots in two noisy views inspired by re-corruption [25]:

$$Z_+ = |Z + \eta|, \quad Z_- = |Z - \eta|, \qquad \eta_{np} \sim \mathcal{CN}(0, v). \quad (20)$$

The estimator receives Z_+ and variance $2v$; the Rician loss uses target Z_- and variance $2v$. Pilots and reference are unchanged. The views use measured magnitudes, not recovered complex fields or true channels. Circular Gaussian field noise gives the required independent magnitude-view law; the supplement gives the argument and its assumptions. At test time, both recipes use all original measurements and their supplied variance. This comparison changes training noise, pilot use and the loss together, so it does not isolate the effect of pilot count alone.

C. Matched Compact Controls

The step-only control learns ten update parameters and does not filter. Global filtering adds one threshold and gate per stage, giving 30 parameters. Both share our initialization and EM core; global filtering also shares the smooth operation.

The NA-ALISTA-inspired control feeds two observed quantities to a shared long short-term memory (LSTM) cell with four hidden units: a normalized log amplitude residual and a per-user normalized log squared EM-update displacement. Its hidden and memory states start at zero for every block and user, passing only between stages. A two-output linear map chooses the same positive relative threshold and sigmoid mixing as our controller. The LSTM has 128 parameters, the output map has ten and the learned EM steps have ten: 148 total. Initial steps, filtering and mixing match the filtered arms.

This adaptation changes NA-ALISTA's measurement model, features, initial states, readout, controlled quantities and loss. Its EM steps remain learned stage constants, rather than recurrent outputs. We call it inspired, not reproduced. Features and architecture both differ from our controller, so the comparison cannot attribute a gain solely to spectral inputs or recurrence. Exact equations and implementation tests are supplied in the supplement.

D. Properties of Individual Updates

Define $F_0(H) = H$ by continuity. For fixed $\tau > 0$, each multiplier in (13) lies in $[0, 1]$. Therefore

$$\|F_\tau(H)\|_F \le \|H\|_F, \qquad \|F_\tau(H) - H\|_F \le \sqrt{q}\,\tau/2. \quad (21)$$

The second result follows by maximizing $\sigma\tau^2/(\sigma^2+\tau^2)$, whose maximum is $\tau/2$. Let D_H contain antidiagonal multiplicities, so $\|g\|_{D_H} = \|\mathcal{H}(g)\|_F$. Antidiagonal averaging is the lifted-space orthogonal projection onto Hankel matrices. The gated step thus does not increase this weighted norm and has weighted correction at most $\beta\sqrt{q}\tau/2$. This bounds the step's output, not its error relative to the true channel or contraction between two inputs.

The EM surrogate is $\|Q - B - GS\|_F^2/v$. Its normal equations show that relaxation reduces this surrogate by

$$\gamma_t(2-\gamma_t)\|(G - G_{\text{EM}})S\|_F^2/v \ge 0 \quad (22)$$

for $0 \le \gamma_t \le 2$. Subsequent filtering need not decrease the surrogate or likelihood. No convergence, fixed-point stability or low-noise consistency theorem follows for the full estimator. In particular, the learned threshold need not vanish with noise. A likelihood safeguard would not itself guarantee smaller channel error and is not added here. Cubic shrinkage remains a design choice, not a statistically optimal filter for the induced Hankel errors.

V. Experimental Design

A. Separate Classical and Learned Protocols

Classical experiments test structural assumptions before introducing learning. The main array study uses $K = 3$, $N = 8, 16, 32$, $P = 10, 30$, RSR 12 dB, six SNRs from -5 to 20 dB, and 400 paired worlds per cell. Unless fixed, each user's path count is independently uniform on 3–7. The generator rejects circular frequency separation below 10^{-10}, steering-matrix singular-value ratio below 10^{-8} and pilot singular-value ratio below 10^{-8}. Results apply to these screened samples.

Effective-rank tests use $N = 16, 32, 64$, $P = 20$, RSR 10 dB, random SNR over $[-10, 20]$ dB, and 300–400 worlds per cell. A filter comparison uses $N = 32$, $P = 30$, RSR 12 dB, $L = 3, 7, 14$, SNR 0 and 10 dB, and

400 worlds per cell. A separate fresh sweep check uses 100 per cell. These studies are not pooled with learned results.

All compact arms use 16,384 training blocks, 2,048 validation blocks, Adam with learning rate 10^{-3}, batch size 32, 20 epochs and seeds 9101–9103. Training uses single precision; testing uses double precision. Half the training blocks draw 3–7 paths per user. The remainder draw a shared count from $\{2, 4, 8, 12, 16\}$. Seven SNRs from -10 to 20 dB and RSRs 5,12,20 dB define 126 validation conditions.

The compact study reuses the original selected step-only, global and 188-parameter checkpoints and trains the recurrent arm with matched data, masks, batch orders and selection. Its selected epochs are 20,19,20, with late validation improvement. A matched budget is not convergence. All three seeds share one training dataset, so they do not measure training-population uncertainty.

B. Fresh Paired Comparison

The main final test crosses seven SNRs, three RSRs and six path regimes: independent 3–7 paths or fixed counts 2,4,8,12,16. It uses $N = 32$, $K = 3$, $P = 30$. Each of two fresh banks has 16 worlds per cell, giving 4,032 primary worlds. Two separate cells add 64 stress worlds: SNR 5/RSR 0 dB and SNR 60/RSR 12 dB, both with 3–7 paths. Stress cells do not enter the primary average.

Every world redraws channels, pilots and noise, paired across methods. Sources, data and checkpoint hashes are frozen before test generation. Neural seeds are neither selected nor ensembled. OptShrink settings come from the earlier validation-only search. Each follow-up uses a separate fresh bank; results from different studies are not pooled.

The larger all-pilot comparison uses the same training examples, initializations, orders and update budget as the original models. Both recipes retain the common 21/9 validation rule, which can favor the original recipe; final-epoch comparisons are reported separately. Two fresh banks give 8,064 primary and 128 stress worlds. Mixed training uses one seed and 4,096 new worlds. The selector uses one seed and 4,224 new worlds, including 192 stress cases. The latter two studies are development screens, not independent confirmation of a replacement method.

C. Metrics, Uncertainty and Processing Cost

For method a and world i, let $e_{a,i} = \|\widehat{G}_{a,i} - G_i\|_F^2$ and $d_i = \|G_i\|_F^2$. Cell normalized mean-square error (NMSE) is

$$E_a = 10 \log_{10} \frac{\sum_i e_{a,i}}{\sum_i d_i}. \tag{23}$$

The primary compact score averages equally over 126 cell dB scores and then over three training seeds. This is not an ensemble prediction. Classical scores use the same cell rule. For classical diagnostics, we also use the paired median $\mathrm{median}_i[10 \log_{10}(e_{a,i}/e_{b,i})]$, which need not equal $E_a - E_b$. Positive gain favors method b.

Three predeclared compact contrasts compare our estimator with the step-only, global and recurrent controls. We use 4,000 paired bootstrap resamples within bank and cell, keeping all methods and seeds paired. Bonferroni-adjusted two-sided intervals have nominal familywise level 95%. They condition on fitted models and development data. Classical differences on this final bank are descriptive; separate earlier studies report their own adjusted intervals. We also report pooled linear error and fractions with per-world NMSE above $0.5, 1, 2$. NMSE above one is called severe error, not communication outage.

Final timing uses one Apple M2 CPU thread, PyTorch for every method, double precision and batch size one. An independent world per cell is measured after warmup with three randomized-order repeats. Initialization, features, structural operations, rank search and all-pilot refitting are included. Loading, file access, offline tuning and training are excluded for every method. Older mixed-backend times are kept separate. These are implementation-specific costs, not device-independent speed claims.

VI. Results

A. Array Size and Useful Structure

At $N = 32$, $P = 30$, RSR 12 dB, HS-amplitude gains 2.54–2.97 dB in paired median over EM-GS across six SNRs, averaging 2.70 dB (Fig. 2). Ratio-of-sums gaps are 2.38–2.90 dB. At SNR at least 5 dB, EM-GS is within 0.108 dB of the Rician CRLB. The geometric CCRB is 6.95–7.06 dB below that reference, while HS-amplitude remains 4.37–4.85 dB above it. These local unbiasedness references do not establish efficiency or attainable headroom for the biased filtered estimates.

Averaging two pilot counts and six SNRs, paired-median gains are 0.236, 1.044 and 3.264 dB for $N = 8, 16, 32$; ratio-of-sums gains are -0.095, 0.866 and 2.938 dB. Larger arrays provide more spatial samples of the same paths. At $N = 8$, maximum Hankel rank is four and filtering is bypassed in 42.94% of trials. A positive median alongside a negative total-error gain shows why aggregation matters.

B. Effective Rank and Filtering Bias

For the true noiseless Hankel spectrum, define effective rank [26]

$$r_{\mathrm{eff}} = \exp(-\textstyle\sum_i p_i \log p_i), \qquad \rho = r_{\mathrm{eff}}/q. \tag{24}$$

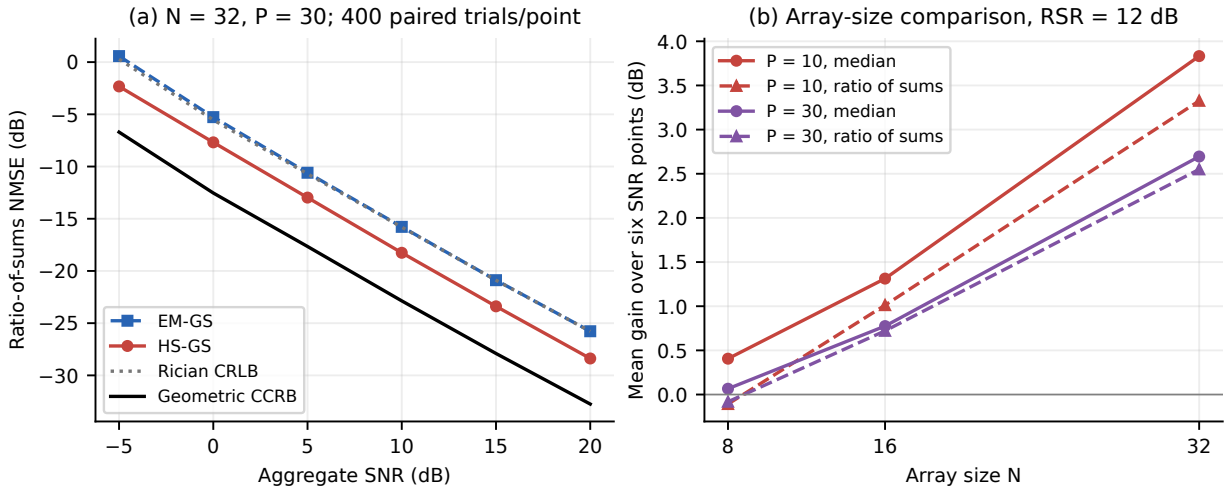


Fig. 2. Classical study. Left: NMSE and information references at $N = 32$, $P = 30$, RSR 12 dB. Right: HS-amplitude gain versus array size for two pilot counts. HS-GS denotes four-sweep HS-amplitude. Solid gain curves average paired medians over six SNRs; dashed curves average ratio-of-sums gains.

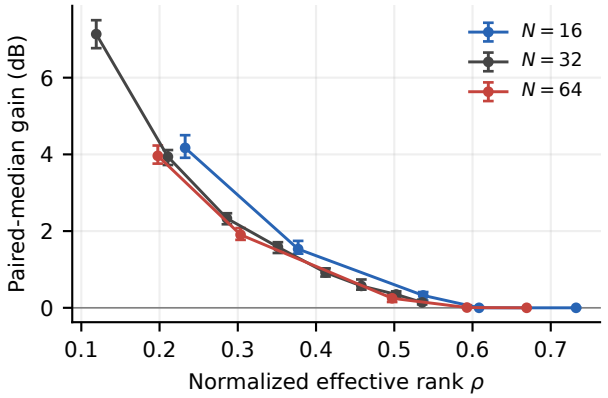


Fig. 3. Four-sweep HS-amplitude paired-median gain versus true-channel normalized effective rank. Arrays use $P = 20$, RSR 10 dB and the same random-SNR distribution. Bars are pointwise paired-bootstrap intervals. This is a diagnostic trend, not an observable switching threshold.

TABLE I
COMPACT-CONTROLLER COMPARISON: 4,032 FRESH PRIMARY WORLDS. LOWER NMSE IS BETTER. NEURAL SCORES AVERAGE THREE MODELS, NOT PREDICTIONS. TIME INCLUDES INITIALIZATION AND RANK-SEARCH/REFITTING WHERE USED.

Method	Weights	NMSE (dB)	Time (ms)	Severe (%)
Learned steps	10	−9.773	1.79	22.84
Global filtering	30	−11.857	3.41	19.56
Recurrent adaptation	148	−11.725	4.15	19.78
Our spectral/state control	188	−13.692	4.87	3.10
OptShrink, offline	—	−13.204	15.81	4.46
OptShrink, block rank	—	−14.144	110.56	6.77

A spectrum dominated by a few values has low effective rank even if its algebraic rank is larger. We report median ρ over users and worlds. This true-channel diagnostic is not available to the estimator. Across the matched array tests, hard-rank benefit generally falls as ρ rises (Fig. 3). At $N = 32$, $\rho = 0.536$ gives median gain 0.141 dB with interval $[0.090, 0.211]$ dB. A value near one-half of capacity marks a small benefit here, not a universal threshold. At $N = 16$, $L = 14$, $\rho = 0.732$, forcing rank seven loses 0.154 dB, with interval $[0.111, 0.201]$ dB for the loss.

Adaptive rank does not protect every channel: at $N = 32$, $L = 14$, SNR 0 dB, HS-amplitude is worse than EM-GS in 47.75% of 400 worlds. Selecting below the physical path count is not always harmful either. At $L = 7$, SNR 0 dB, 78.75% select below rank seven, but only 8.75% of all worlds are worse than EM-GS.

Classical smooth filtering gains 0.914, 1.382 and 1.413 dB over four hard sweeps at SNR 0 dB for $L = 3, 7, 14$. At 10 dB, gains are 0.005, 0.623 and 0.773 dB; the first interval includes zero. Learning is therefore not required to improve hard truncation. A fresh check also finds one hard sweep better than four in all six cells by 0.275–0.652 dB. More filtering is not automatically better.

C. Matched Compact Comparison

Table I compares all six methods. Our controller improves on learned steps alone by 3.919 dB, with adjusted interval $[3.789, 4.001]$ dB, and on global filtering by 1.835 dB, with interval $[1.805, 1.871]$ dB. It improves on the recurrent adaptation by 1.967 dB, with interval $[1.932, 2.006]$ dB. It wins 114 of 126 cells against

TABLE II
MATCHED TRAINING-RECIPE COMPARISON: 8,064 FRESH PRIMARY WORLDS. NEURAL RESULTS AVERAGE THREE SEPARATE FITS, NOT AN ENSEMBLE. ERROR IS THE EQUAL-CELL DB NMSE; SEVERE MEANS PER-CASE NMSE > 1. ONE-THREAD CPU TIME INCLUDES INITIALIZATION AND ANY RANK SEARCH/REFIT.

Method	NMSE (dB)	Severe (%)	Time (ms)
Original	-13.684	2.82	4.81
All-pilot	-13.962	4.47	4.81
OptShrink, offline	-13.168	4.04	15.66
OptShrink, block rank	-14.132	6.76	108.08

TABLE III
SEPARATE FINAL STRESS CELLS: 32 PAIRED WORLDS EACH. NMSE IN DB; LOWER IS BETTER. NEITHER ENTERS SELECTION OR THE PRIMARY AVERAGE.

Method	Weak reference SNR 5/RSR 0 dB	Nearly noiseless SNR 60/RSR 12 dB
Our controller	-6.842	-6.321
Recurrent adaptation	-5.623	-20.101
OptShrink, offline	-10.823	-67.701
OptShrink, block rank	-12.625	-70.336

recurrence, with worst observed cell loss 0.176 dB. This is evidence for the tested design under a matched training budget, not a claim that every recurrent controller or published NA-ALISTA is inferior.

Compared with offline-selected OptShrink, our descriptive average gain is 0.488 dB and mean time is about 3.24 times lower. Per-block in-loop rank selection reverses the accuracy ranking: OptShrink gains 0.452 dB, while our measured time is about 22.7 times lower. Our worst primary-cell loss to that selector is 5.848 dB. The comparison includes classical search and all-pilot refitting, but excludes neural training and classical offline tuning. It represents a reusable trained estimator versus additional per-block computation, not a free accuracy gain.

Figure 4 shows the SNR and cost trends. Original seed scores are -13.784, -13.449 and -13.843 dB; recurrent scores are -11.707, -11.740 and -11.728 dB. The recurrent model is smaller and slightly faster. Its accuracy gap under this budget does not establish its best performance after longer training or different features.

D. Training With All Pilots

The larger, separate comparison in Table II retains the original architecture and OptShrink settings. All-pilot training gains 0.277 dB over the original recipe, with adjusted conditional 95% interval $[0.263, 0.294]$ dB, and wins 106 of 126 cells. Severe errors nevertheless rise from 2.82% to 4.47%. Inference time is unchanged at about 4.81 ms. Rank-selected OptShrink remains more accurate by 0.170 dB, with interval $[0.117, 0.221]$ dB for its advantage, at about 108 ms. Fixed final-epoch comparison still favors all-pilot training over the original by 0.242 dB. The benefit is not solely a selected-checkpoint result, but it is not a reliability improvement or proof of convergence.

A one-seed 50/50 mixture alternates the original and all-pilot training losses, keeping the number of updates fixed. On its separate fresh bank, mixed training gains 0.152 dB over the original. Its severe-error rate is 3.75%, between the original's 2.90% and all-pilot's 4.76%, and time remains about 5 ms. It fails the declared screening targets for average gain and protection of low-SNR errors. We do not replace the original method with this mixture.

A simpler control chooses one of the two frozen models using supplied noise variance. Held-out likelihood selects the constant "always original" rule, so it gives zero accuracy gain and about 1.4% timing overhead. Selection uses the original 21/9 validation split and can favor that recipe; this result does not rule out other selectors. We retain the failed screen rather than choose a new threshold from test errors.

E. Error Aggregation and Earlier Evidence

Our severe-error rate is 3.10%, compared with 19.78% for recurrence, 4.46% for offline OptShrink and 6.77% for per-block rank selection. This threshold is not the primary selection objective or a communication-outage metric. At NMSE 0.5, exceedance rates are 19.43%, 19.12% and 19.27% for our estimator and the two OptShrink controls. Lower severe-error frequency does not mean our error distribution dominates theirs.

Pooling linear errors over primary worlds before taking dB gives -6.664 dB for ours, -6.564 dB for offline OptShrink and -6.228 dB for per-block selection. That descriptive measure favors ours over the primary accuracy winner. It gives more influence to large errors, whereas the declared primary score gives every operating cell equal dB weight. We retain the primary rule rather than choosing the aggregation with the most favorable ranking.

Earlier banks show the same dependence on the comparator: a broad offline-tuning study favors ours by 0.537 dB, while a fresh per-block rank study favors OptShrink by 0.507 dB. Their full results remain separate in the supplement. The tuning population, metric and strongest comparator must be stated together.

F. Failures and Scope of Robustness

Weak reference and nearly noiseless observations expose substantial failures (Table III). At low noise, our filter can erase information that the measurement update already recovers accurately. A separate 32-world

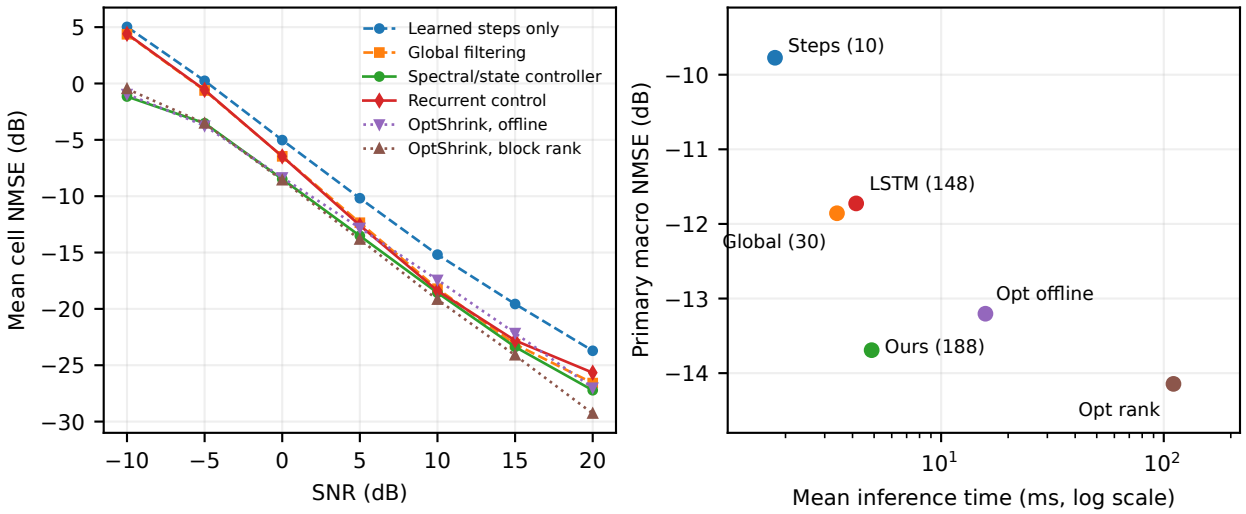


Fig. 4. Compact-controller comparison. Left: each SNR point averages 18 reference/path cells. Right: equal-cell accuracy and serial time. All methods use the same PyTorch CPU backend. Lines are descriptive averages, not simultaneous per-cell confidence guarantees.

diagnostic at SNR 60 dB reaches -6.647 dB with the full model, -44.020 dB when filtering alone is disabled, and -65.715 dB with EM100. Even starting the full learned model at the true channel gives -6.653 dB. These post-hoc tests implicate filtering bias; they do not supply a validated switching rule.

Frozen-model checks vary $N \in \{8, 16, 32, 64\}$, $K \in \{2, 3, 4\}$ and $P \in \{10, 20, 30\}$ one factor at a time. The gain interval at $N = 8$ includes zero. At SNR 5 dB, a smaller study reaches -2.197 dB at $P = 6 = 2K$ versus -14.332 dB at $P = 30$. Local identifiability does not ensure useful finite-noise accuracy.

Noise errors are important in the selector study's 4,032 primary worlds. With correct variance, severe errors are 2.80% for the original and 4.86% for all-pilot training. Supplying variance 3 dB too low raises them to 11.11% and 14.51%; 3 dB too high gives 3.40% and 5.43%. Only the supplied variance changes, not the observations. These settings reuse the same worlds, not three independent populations. The chosen constant selector adds no protection.

Separate reference checks also expose sensitivity. At SNR 5 dB, constant reference-phase errors of $\pm 10^\circ$ worsen learned NMSE by 2.36–2.47 dB. A new oracle phase diagnostic removes substantial weak-reference error by aligning each array row to the true channel, but this uses unavailable truth and is not an estimator. Neither diagnostic solves calibration or demonstrates hardware robustness.

The supplement retains input, depth and placement studies, the small all-pilot screen and negative OptShrink hybrids. None establishes a better replacement for the main method across accuracy, severe errors and cost. A small Transformer pilot is not a matched reproduction of published URformer.

VII. Discussion and Conclusion

Structural filtering must be judged by the noise it removes, the bias it adds and the cost of choosing it. Effective rank explains part of the classical trend, but uses true channels here. A bypass, learned gate or norm bound does not protect every estimate. Stronger classical tuning changes the ranking.

The original estimator is useful when many received blocks can reuse offline training from a known population. It improves on the tested compact controls and offline OptShrink lookup, while per-block rank selection wins the main accuracy average. All-pilot training moves the accuracy–failure trade-off without increasing inference cost; it does not remove it. We keep the original as the main method, not because it is universally best, but because the tested replacements do not preserve its severe-error rate while improving accuracy.

One adaptive stage costs $O(NKP + NK^2 + K^3 + Kmq^2 + Kq^3)$. Features need an SVD and filtering a dense solve. Rank selection repeats this work across candidates. Parameter count is not processing cost. Timing ratios depend on implementation, precision, threads and batching; offline training and tuning also cost resources.

The scope remains simulation with a calibrated reference and known Gaussian-field-noise variance. Mismatch tests reveal weaknesses, not general robustness. Shared training data and finite training budgets limit the confidence intervals. Hardware performance, full-estimator convergence, optimal shrinkage, 2D arrays and published-URformer superiority are not established. Full

LRHD and other low-rank phase-retrieval adaptations remain comparison gaps. The new training tests do not isolate pilot count from training-noise and loss changes.

In conclusion, a small self-supervised controller can learn useful spatial filtering without channel labels or per-block rank search. Classical and learned results identify a conditional accuracy–cost trade-off, not a universally superior filter. Alternative training improves the main average but increases severe errors; mixed training and simple selection do not resolve this. These operating limits are part of the result.

A. Reproducibility

The archive retains each study's protocol and evidence. The compact audit regenerated 4,096 worlds and replayed representative neural and independent classical predictions, with maximum difference 4.40×10^{-13}. Later audits check orders, frozen models, paired cases and headline arithmetic. The selector replays 1,188 predictions with maximum difference 4.83×10^{-15}. Not every historical prediction was replayed. Full arrays and checkpoints are separate; a permanent public code/data destination remains to be assigned.

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